1.2 Sets

In the earlier classes, we have seen that a ***set*** is a collection of well-defined objects. As the theory of sets is the building blocks of modern mathematics, one has to learn the concepts of sets carefully and deeply. Now we look at the term "well-defined" a little more deeply. Consider the two statements:

(i) The collection of all beautiful flowers in Ooty Rose Garden.
(ii) The collection of all old men in Tamilnadu.

The terms "beautiful flowers" and "old men" are not well-defined. We cannot define the term "beautiful flower" in a sharp way as there is no concrete definition for beauty because the concept of beauty varies from person to person, content to content and object to object. We should not consider statements like "the collection of all beautiful flowers in Ooty Rose Garden" as a set. Now, can we say "the collection of all red flowers in Ooty Rose Garden" a set? The answer is "yes".

One may consider a person of age 60 as old and others may not agree. There is no specific and concrete definition for "old men". The second statement can be made more sharply as

"the collection of all men in Tamilnadu of age greater than 70".

Now, the above collection becomes a set because of definiteness in the age. Thus, the description of a set should enable us to concretely decide whether a given particular object (element) is available in the collection or not. So set is a distinguishable collection of objects.

We have also seen and learnt to use symbols like $\in$, $\subset$, $\subseteq$, $\cup$ and $\cap$. Let us start with the question:

"If A and B are two sets, is it meaningful to write $A \in B$?".

At the first sight one may hurry to say that this is always meaningless by telling, "the symbol $\in$ should be used between an element and a set and it should not be used between two sets". The first part of the statement is true whereas the second part is not true. For example, if $A = \{1, 2\}$ and $B = \{1, \{1, 2\}, 3, 4\}$, then $A \in B$. In this section we shall discuss the meaning of such symbols more deeply.

As we learnt in the earlier classes the set containing no elements is called an ***empty set*** or a ***void set***. It is usually denoted by $\emptyset$ or $\{\ \}$. By $A \subseteq B$, we mean every element of the set A is an element of the set B. In this case, we say A is a ***subset*** of B and B is a ***super set*** of A. For any two sets A and B, if $A \subseteq B$ and $B \subseteq A$, then the two sets are ***equal***. For any set A, the empty set $\emptyset$ and the set A are always subsets of A. These two subsets are called ***trivial subsets***. Further, we say A is a ***proper subset*** of B if A is a subset of B and $A \neq B$. That is, B contains all elements of A and at least one element which is not in A. Note that, as every element of A is an element of A, we have $A \subseteq A$. Thus, any set is a subset of itself. This subset is called an ***improper subset***. In other words, for any set A, A is the improper subset of A. It is known that, $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, where $\mathbb{N}$ denotes the set of all natural numbers or positive integers, $\mathbb{W}$ denotes the set of all non-negative integers, $\mathbb{Z}$ denotes the set of all integers, $\mathbb{Q}$ denotes the set of all rational numbers and $\mathbb{R}$ denotes the set of all real numbers. Note that, the set of all irrational numbers is a subset of $\mathbb{R}$ but not a subset of any other set mentioned above.

We learnt that the ***union*** of two sets A and B is denoted by $A \cup B$ and is defined as

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

and the ***intersection*** as

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Two sets A and B are ***disjoint*** if they do not have any common element. That is, A and B are disjoint if $A \cap B = \emptyset$.